RAJALAKSHMI ENGINEERING COLLEGE

Ruchiitha KJ 250701613-2025-CSE

P2

Started on	Thursday, 6 August 2026, 10:08 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:11 AM
Time taken	19 days 1 hour
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int count = 0;

void function(int n)
{
    int c = 0;
    count++;                    /* int c = 0;  -> executed once  */

    int i = n / 2;
    while (1) {
        count++;                /* for(i=n/2;i<n;i++)  -> one step per tes
        if (!(i < n)) break;

        int j = 1;
        while (1) {
            count++;            /* for(j=1;j<n;j=2*j)  -> one step per tes
            if (!(j < n)) break;

            int k = 1;
            while (1) {
                count++;        /* for(k=1;k<n;k=2*k)  -> one step per tes
                if (!(k < n)) break;

                c++;
                count++;        /* c++;   -> one step per execution */

                k = k * 2;
            }
            j = 2 * j;
        }
        i++;
    }

    printf("%d\n", count);
}

int main()
{
    int n;
    scanf("%d", &n);
    count = 0;
    function(n);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course